RATAN TATA MAHARASHTRA STATE SKILLS UNIVERSITY

School of Science Engineering and Technology Skills

Department of Engineering

Subject – AWS Lab

Experiment – 3

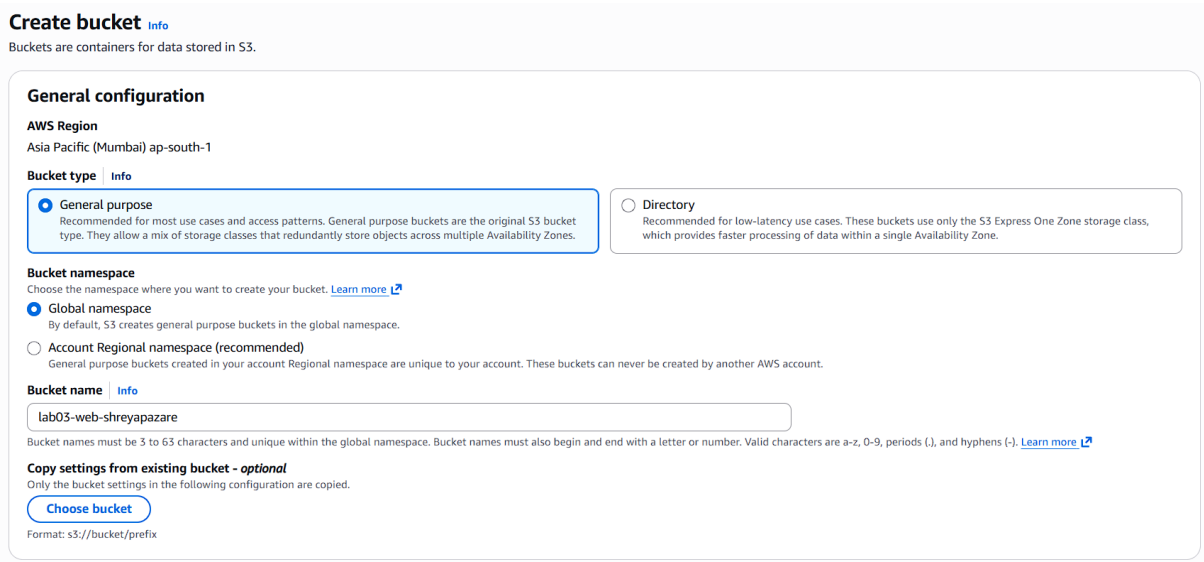
Aim: Amazon S3 & Static Web Hosting

Lab Objective:

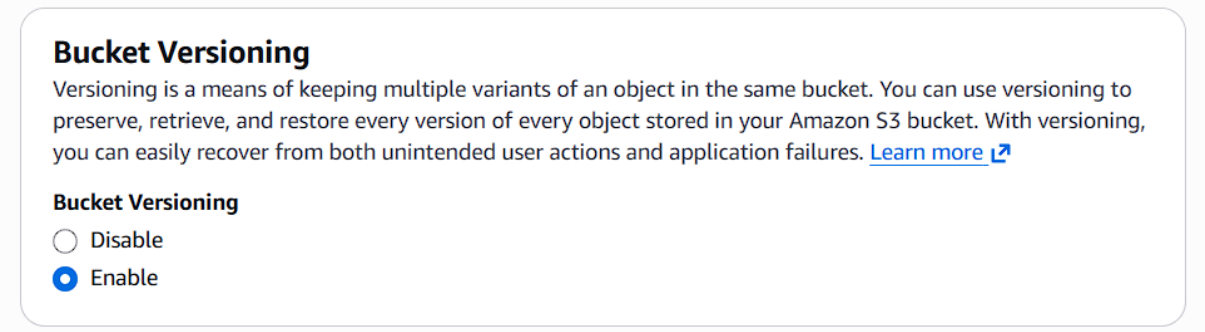
To configure an S3 bucket with advanced features like versioning and encryption, automate cost management via lifecycle policies, and deploy a publicly accessible static website.

Step 1: Create & Secure Your Bucket

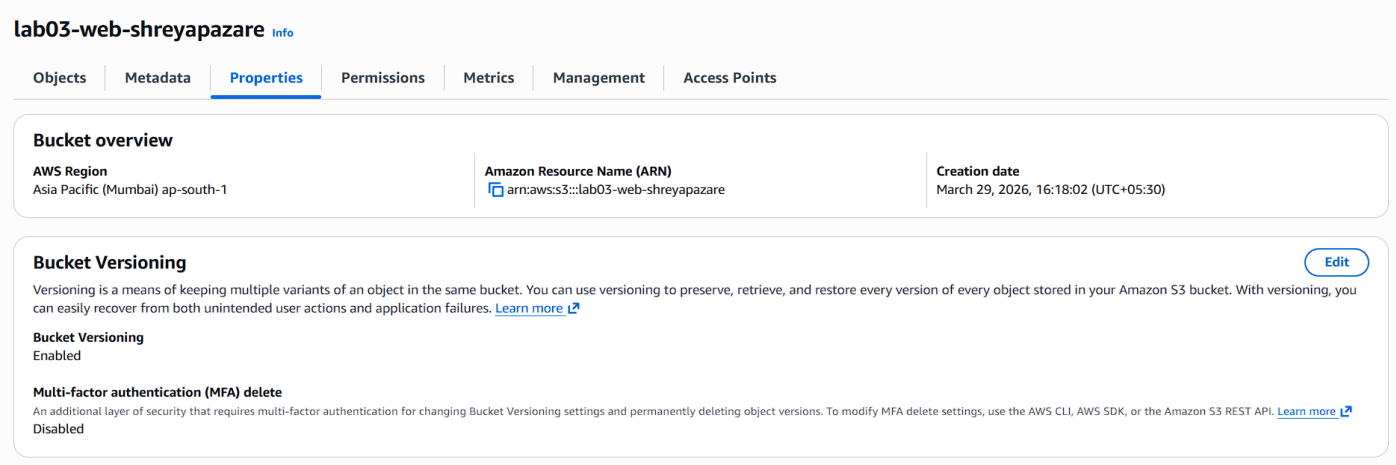
1. Navigate to the S3 Console > Create bucket.
2. Bucket name: lab03-web-yourname (Must be globally unique).
3. Region: Select your closest region (e.g., ap-south-1 for Mumbai



4. Bucket Versioning: Select Enable.
   Why? This protects you against accidental deletes. If you delete a file, S3 just adds a "Delete Marker" and keeps the previous version safe.

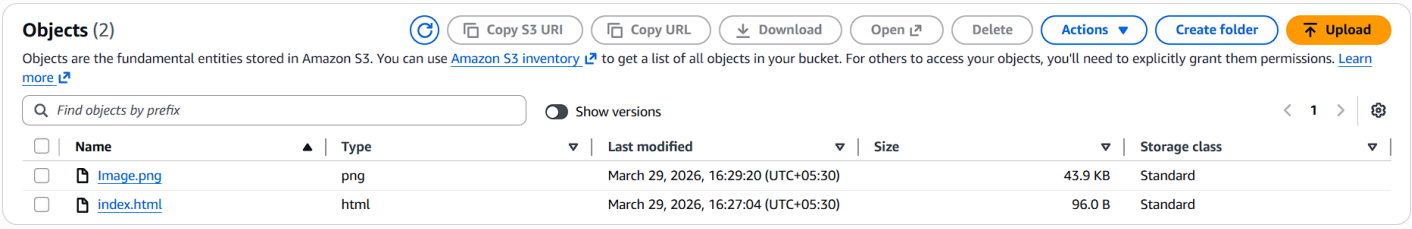


5. Default Encryption: Ensure Server-side encryption with Amazon S3 managed keys (SSE-S3) is enabled (this is now the 2026 standard for all buckets).
6. Click Create bucket.

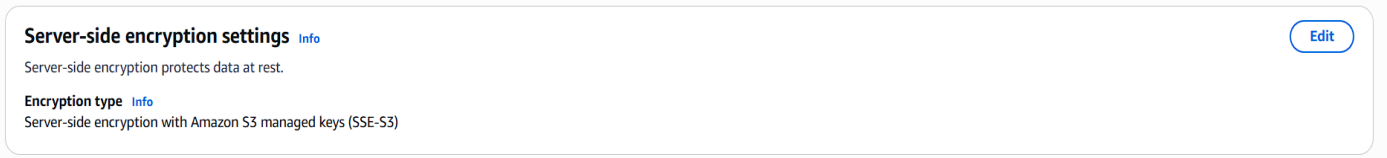


Step 2: Upload & Manage Objects

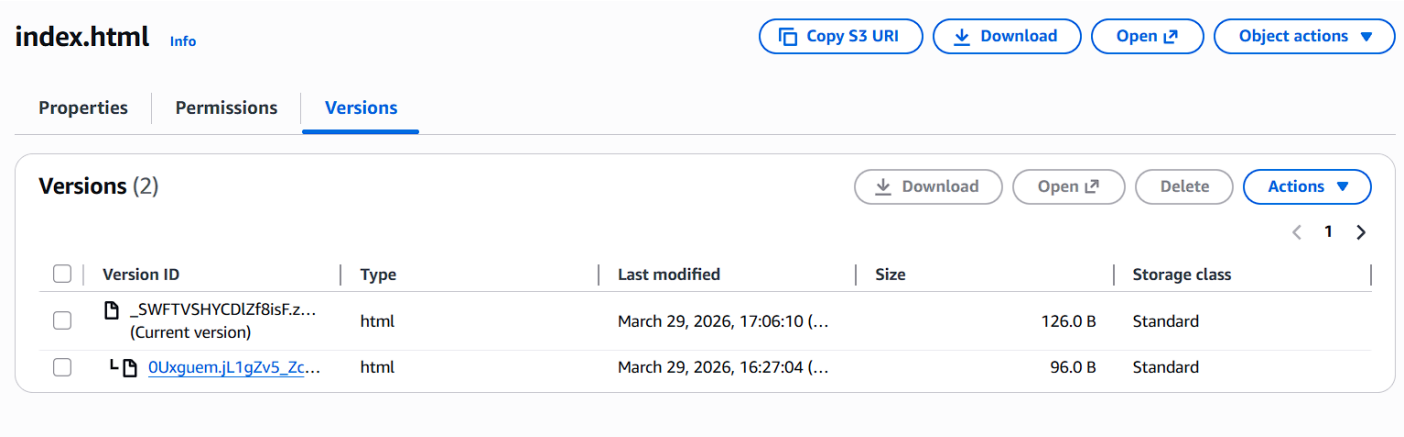
1. Open your bucket and click Upload.
2. Add a simple index.html file (see code below) and an image.



3. Encryption Check: After uploading, click on index.html and scroll to the Properties tab. You will see it is automatically encrypted at rest.



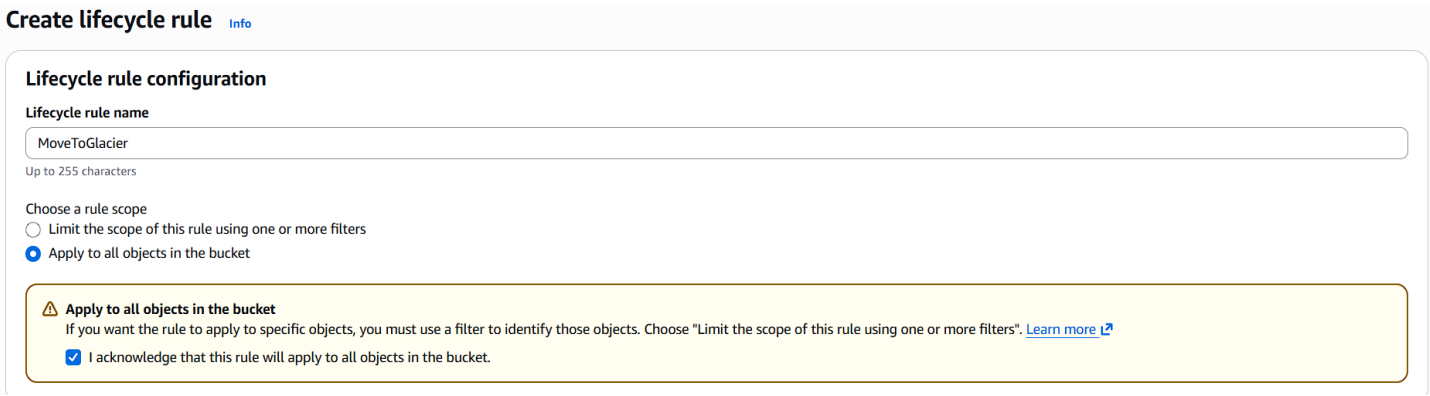
4. Version Test: Upload a different version of the same index.html. Click the Show versions toggle in the bucket objects list to see both timestamps.



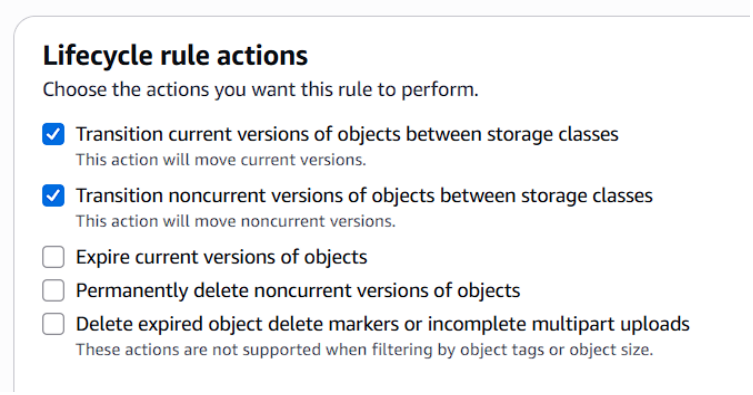
Step 3: Configure S3 Lifecycle Policies

Storage costs money. Use policies to move old data to cheaper tiers automatically.

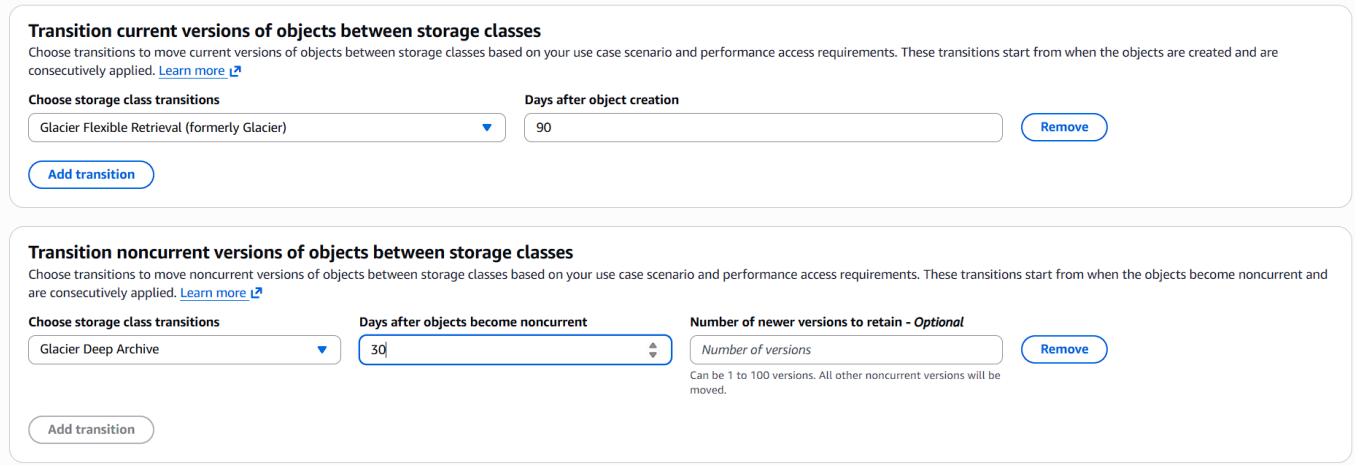
1. Inside your bucket, go to the Management tab.
2. Click Create lifecycle rule.
3. Rule name: MoveToGlacier.
4. Rule scope: Choose "Apply to all objects in the bucket."

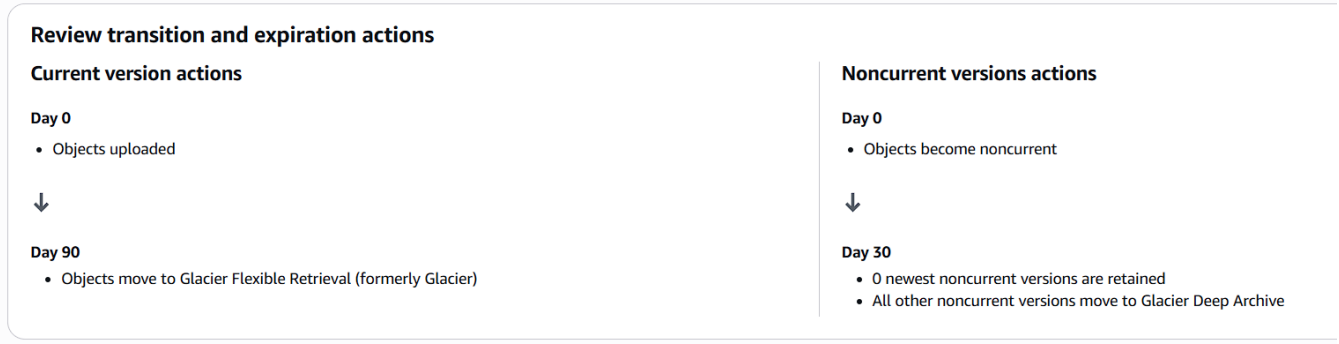


5. Lifecycle rule actions:
    - Select Move current versions of objects between storage classes.
    - Select Move noncurrent versions of objects between storage classes.

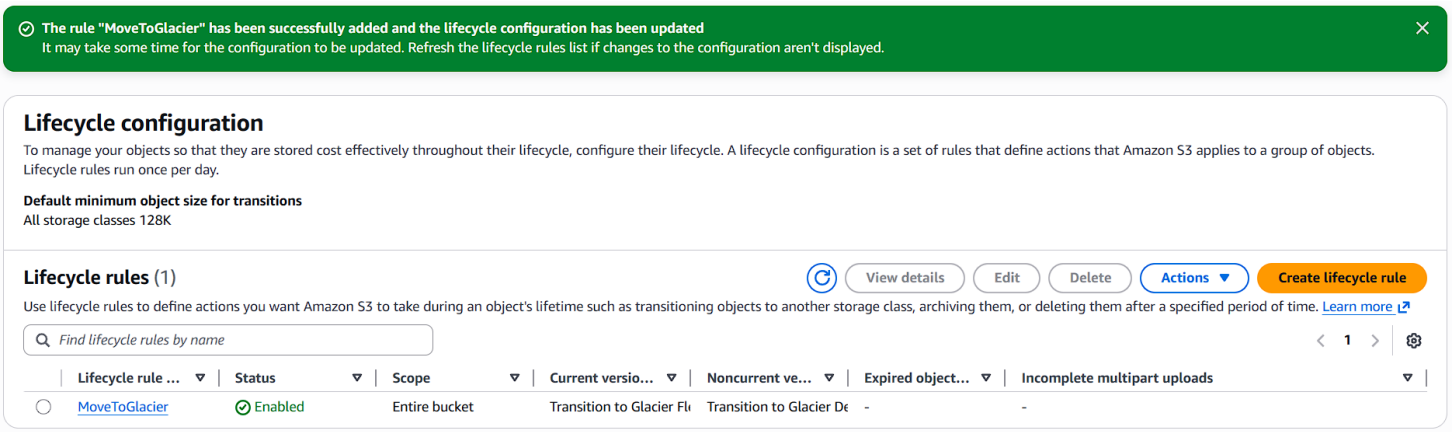


6. Transitions:
    - Transition current versions to S3 Glacier Flexible Retrieval after 90 days.
    - Transition noncurrent versions to S3 Glacier Deep Archive after 30 days.





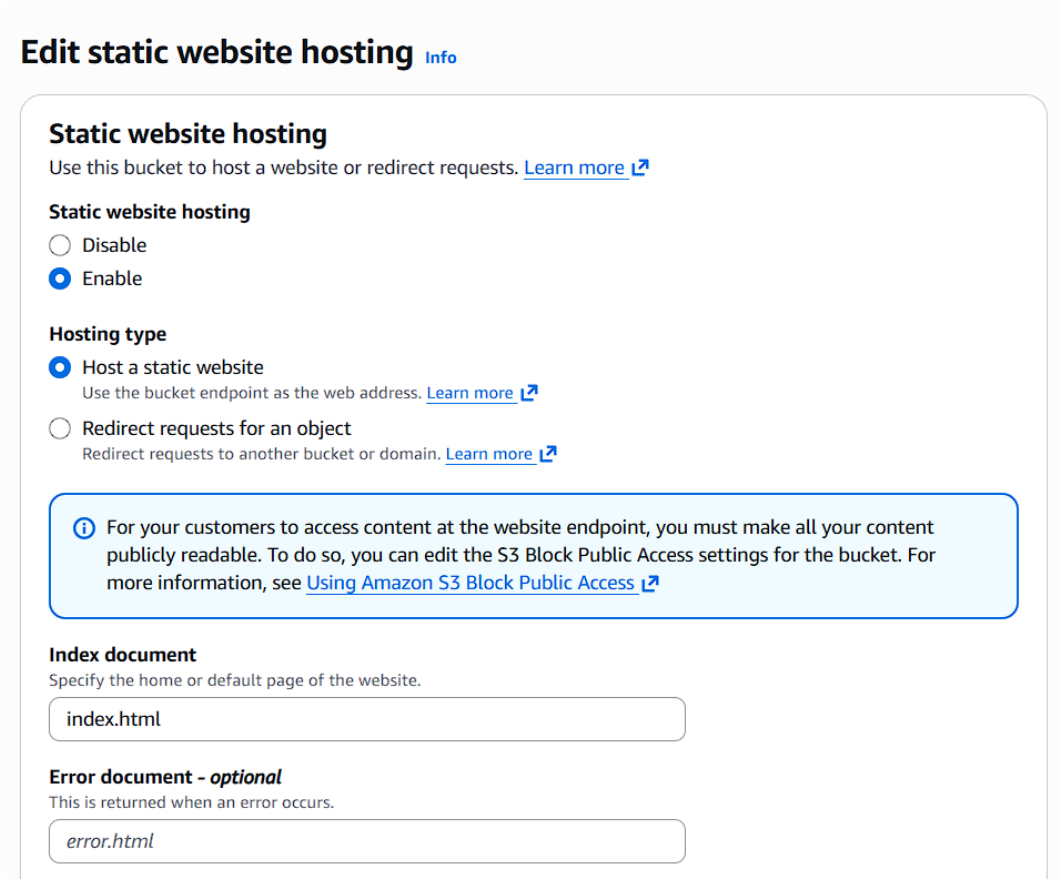
7. Click Create rule.

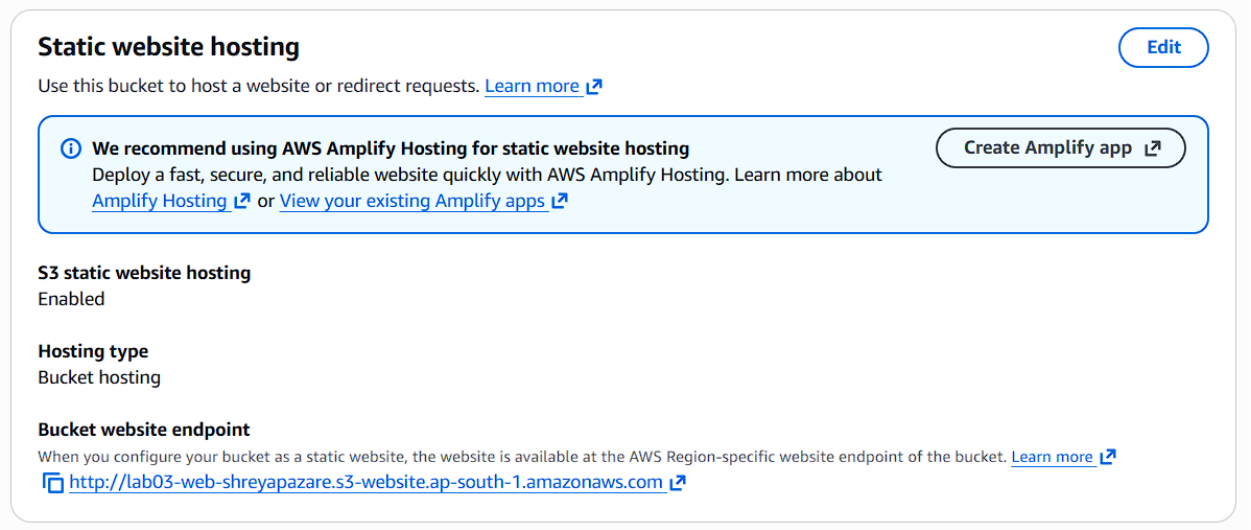


Step 4: Configure Static Website Hosting

1. Enable Hosting

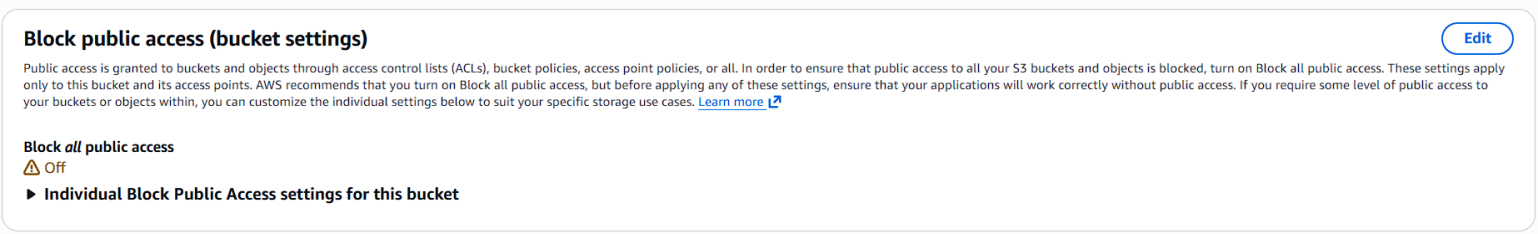
1. Go to the Properties tab > Static website hosting (at the bottom) > Edit.
2. Select Enable.
3. Index document: Type index.html.
4. Click Save changes. Note the "Bucket website endpoint" URL provided.





2. Disable "Block Public Access"

1. Go to the Permissions tab.
2. Under Block public access, click Edit.
3. Uncheck "Block all public access" and click Save. Type confirm.



3. Add a Bucket Policy

The bucket is now "not blocking" public access, but we still need to explicitly Allow it.

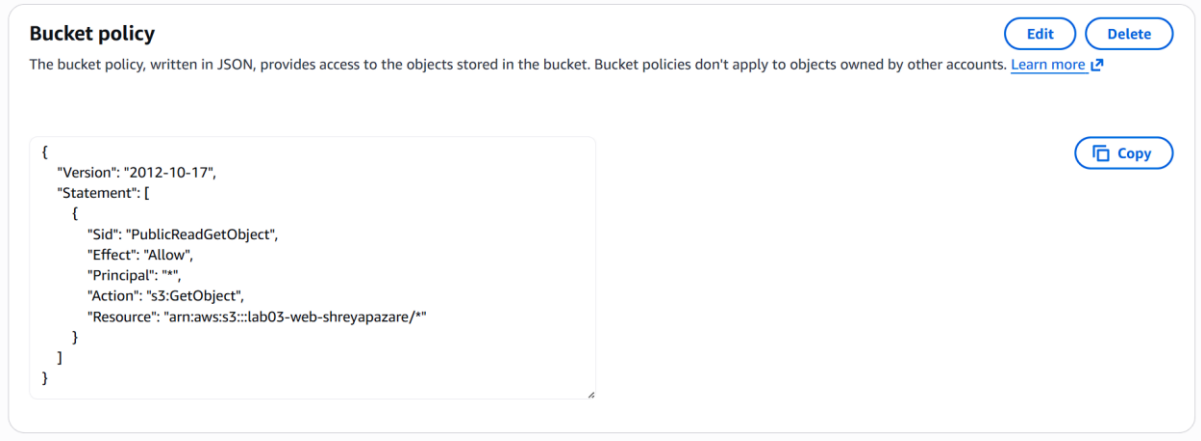
1. In the Permissions tab, scroll to Bucket policy > Edit.
2. Paste the following (replace YOUR-BUCKET-NAME with yours):

JSON:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "PublicReadGetObject",
      "Effect": "Allow",
```

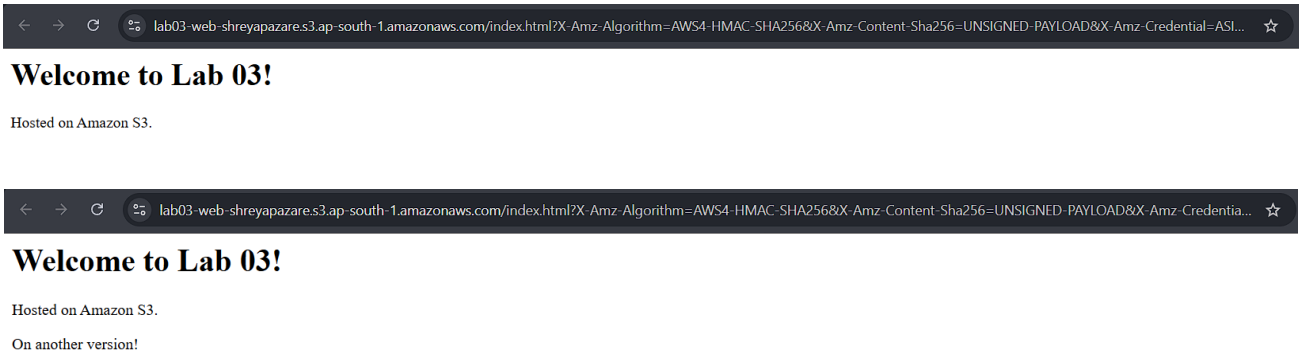
```
            "Principal": "*",
            "Action": "s3:GetObject",
            "Resource": "arn:aws:s3:::YOUR-BUCKET-NAME/*"
        }
    ]
}
```

3. Click Save changes.



Step 5: Verification

1. Create a local file named index.html:
   HTML:
   <html><body><h1>Welcome to Lab 03!</h1> Hosted on Amazon S3. </body></html>
2. Upload it to your bucket.
3. Click the Bucket website endpoint from the Properties tab. You should see your website live!



S3 Security Comparison: IAM vs. Bucket Policies

Feature	IAM Policy	Bucket Policy
Attached To	Users or Roles.	The Bucket itself.
Best For	Managing what one user can do across many buckets.	Managing who (including anonymous public) can access a specific bucket.
Max Size	Smaller (limited by IAM quotas).	Larger (up to 20 KB).